PRE-PRESS TECHNOLOGIES

GRAPHIC DESIGN & MULTIMEDIA PROGRAM

COURSE CODE: GDM5310

ACADEMIC YEAR: 2026/2027

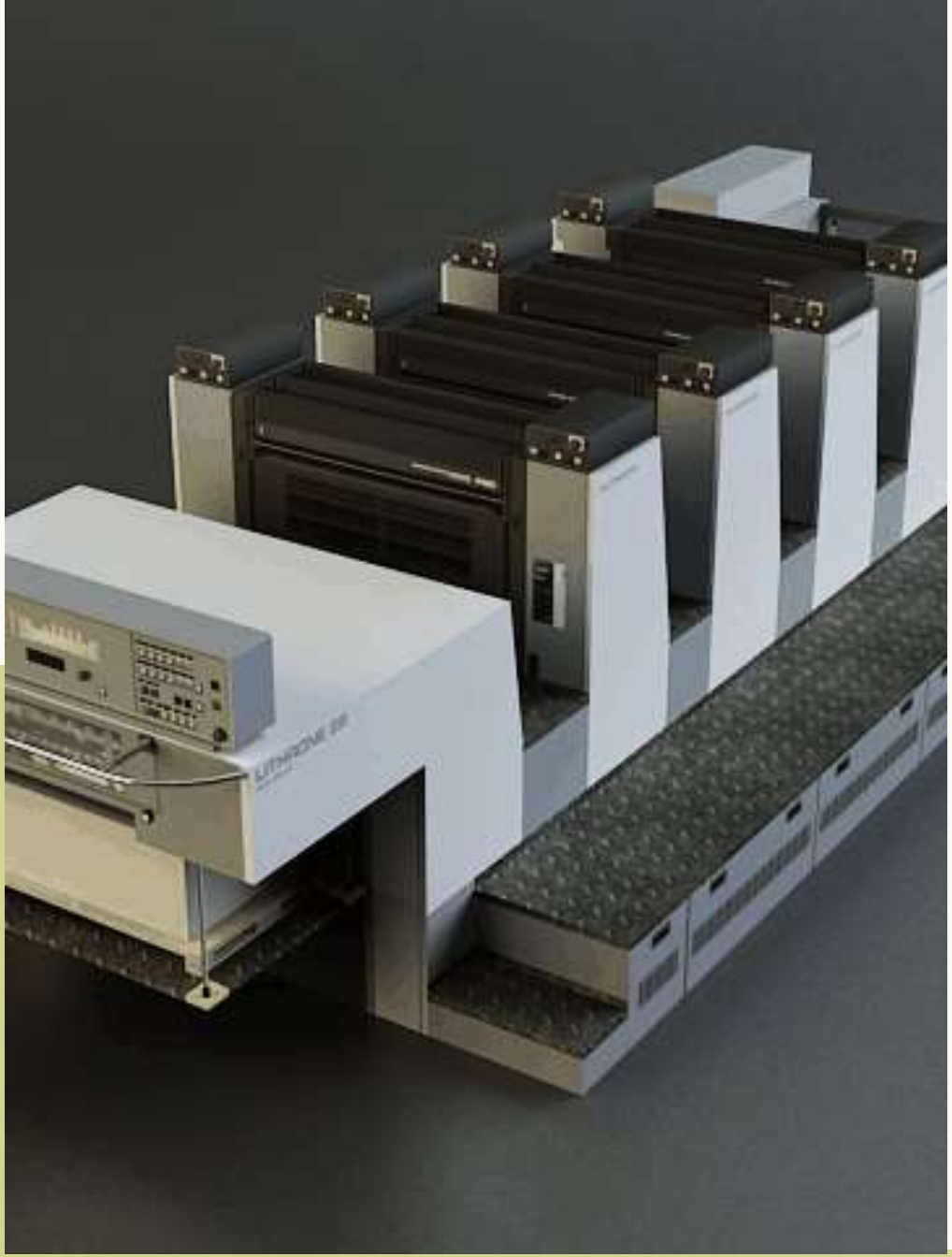
LEVEL: 3

LECTURE 1

BY: DR/ RANIA MOHAMED

AGENDA

- Course aim
- Introduction to Printing Techniques

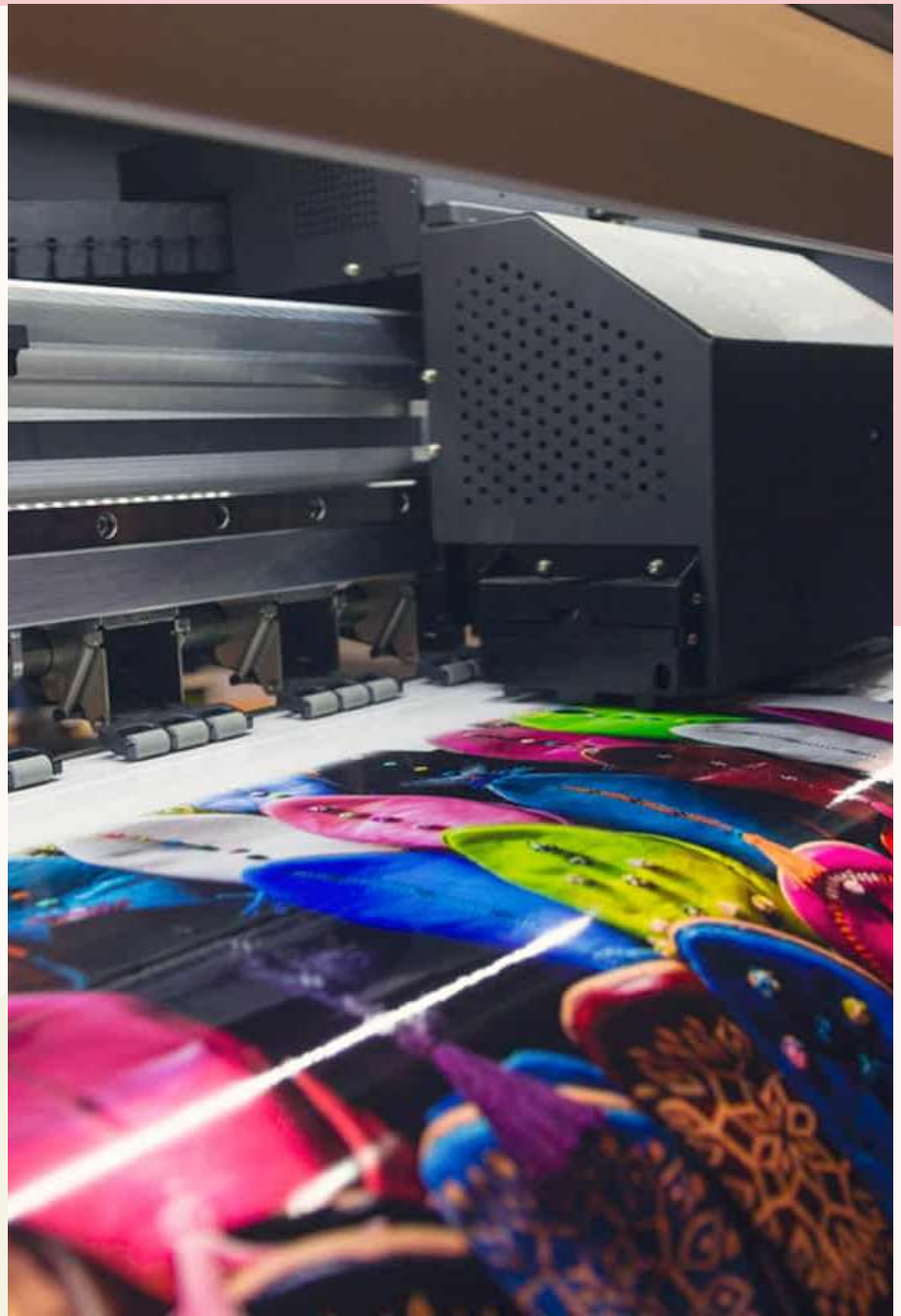


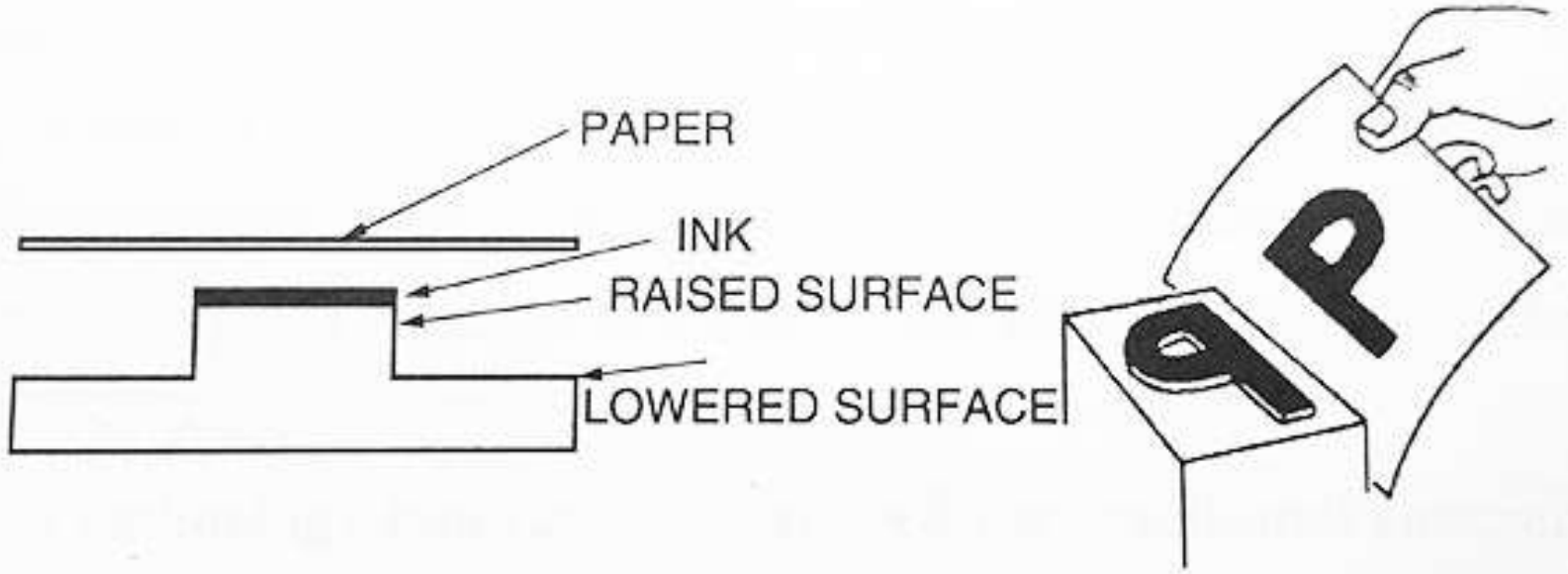
COURSE AIM

Digital systems for pre-press operations to produce various printing surfaces using different lasers, hardware operating requirements, and application programs. The student also learns about the different types of plate production techniques using computers and electronic editing programs.

INTRODUCTION TO PRINTING TECHNIQUES

- Printing is the process of creating physical copies of digital or physical content by applying ink, toner, or other materials to a surface, such as paper, fabric, or metal.
- Key historical developments include Johannes Gutenberg's printing press. Modern printing encompasses diverse techniques like inkjet, laser, offset, flexography, and 3D printing, used for everything from personal documents to large-scale commercial production and manufacturing.





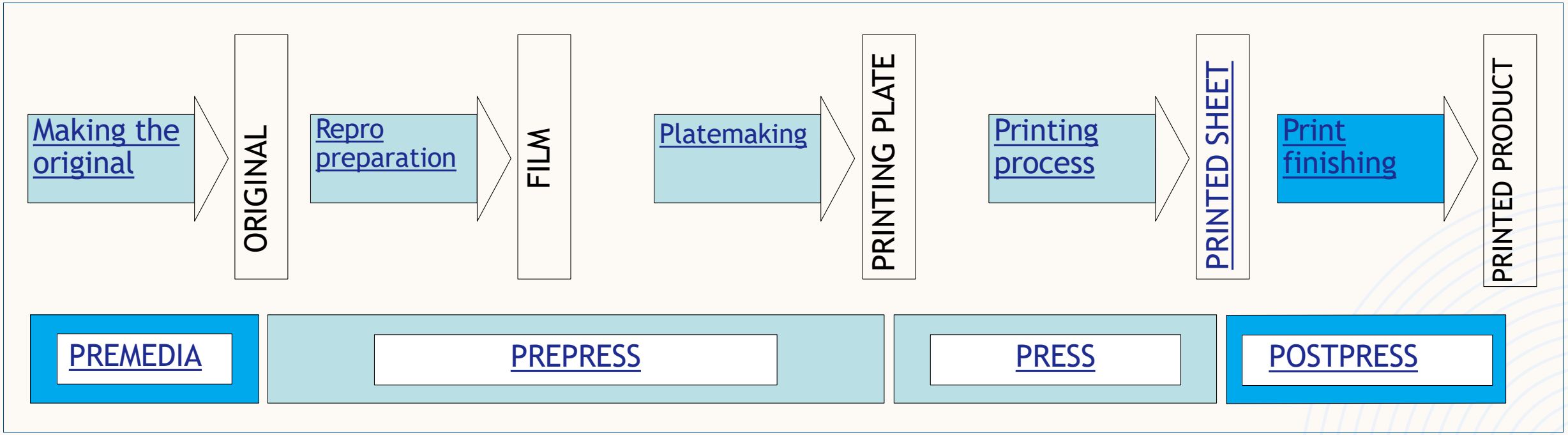
The process of producing printed products can be divided into three parts:

Prepress → Print → Post-Press

- Printed products are graphic products in the widest sense: books and brochures, newspapers and magazines, leaflets and posters, labels and stickers, stamps and securities, credit and check cards, as well as folding cartons and blister packs, bags, paper tissues, and much more.

Prepress includes all the processing steps required to go from the preparation of the text, original images and graphics as well as the design concept right up to the production of the ready-for-print master; the master (printing plate) is then employed at the printing stage to produce the job.

In a wider sense, therefore, “prepress” is also referred to as “pre-media”, meaning the preparation of text and image for the several publication media. As can be seen in the figure, the premedia production stage may already be incorporated in the actual prepress stage, conventional and digital are discussed.



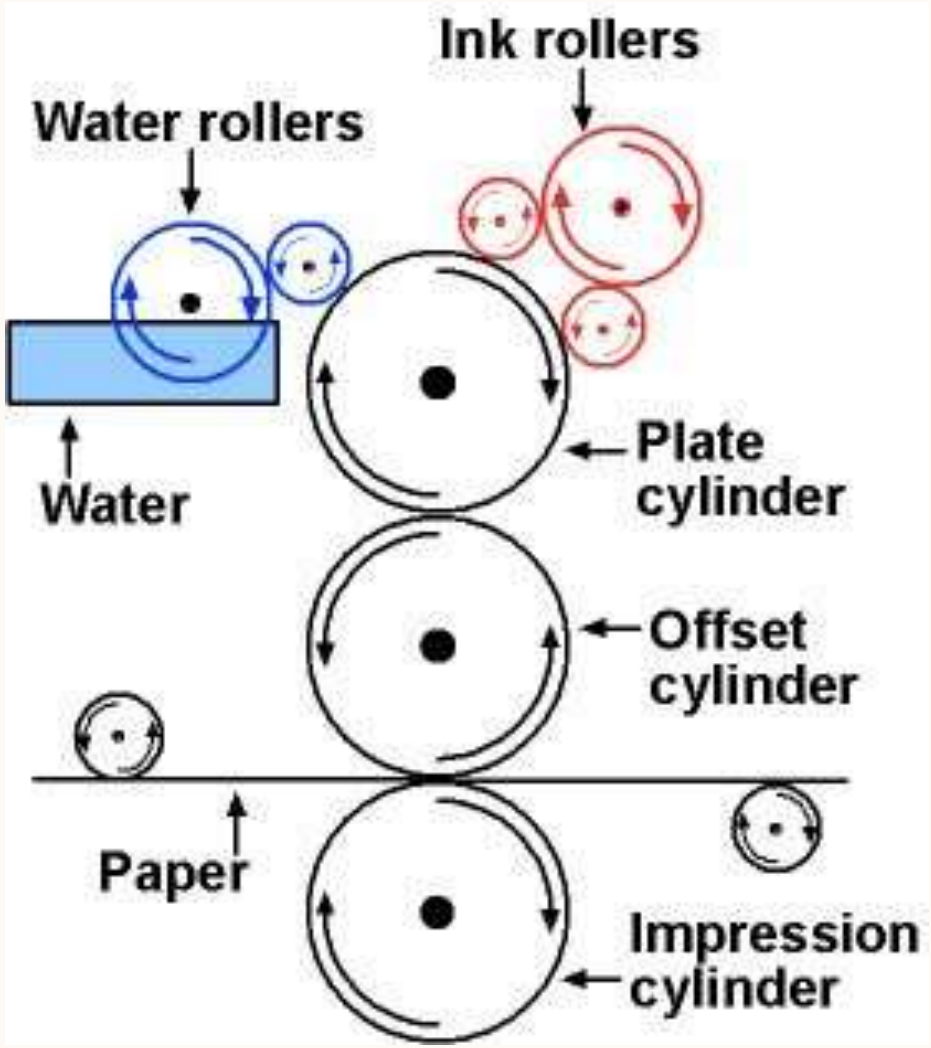
- The distinguishing factor between them is the way the films are used: all technological variations where individual films are used for the page or print sheet composition and require conventional, mechanical /manual film assembly processes, and do not therefore employ digital full-page film exposure systems, are dealt with under the category “conventional prepress.” The section “digital prepress” outlines the processes of digital image processing, computer to plate, computer to film, and so on.

OFFSET PRINTING

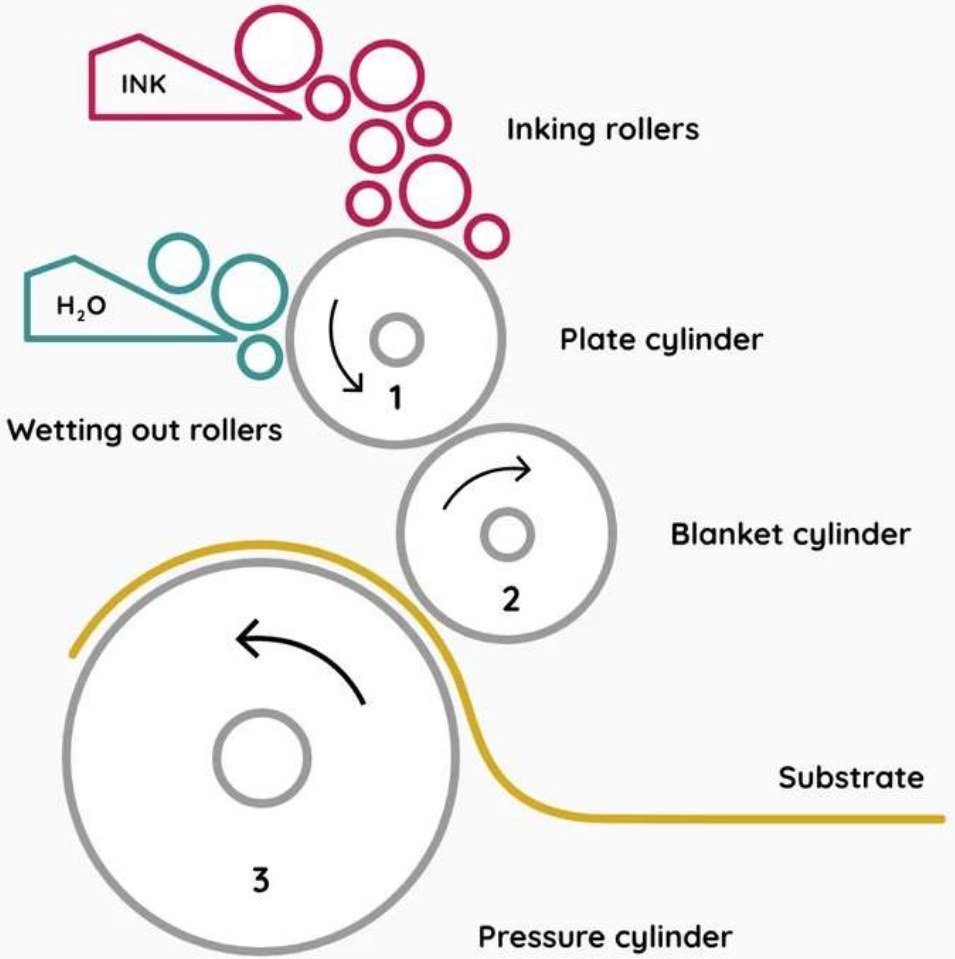


OFFSET PRINTING

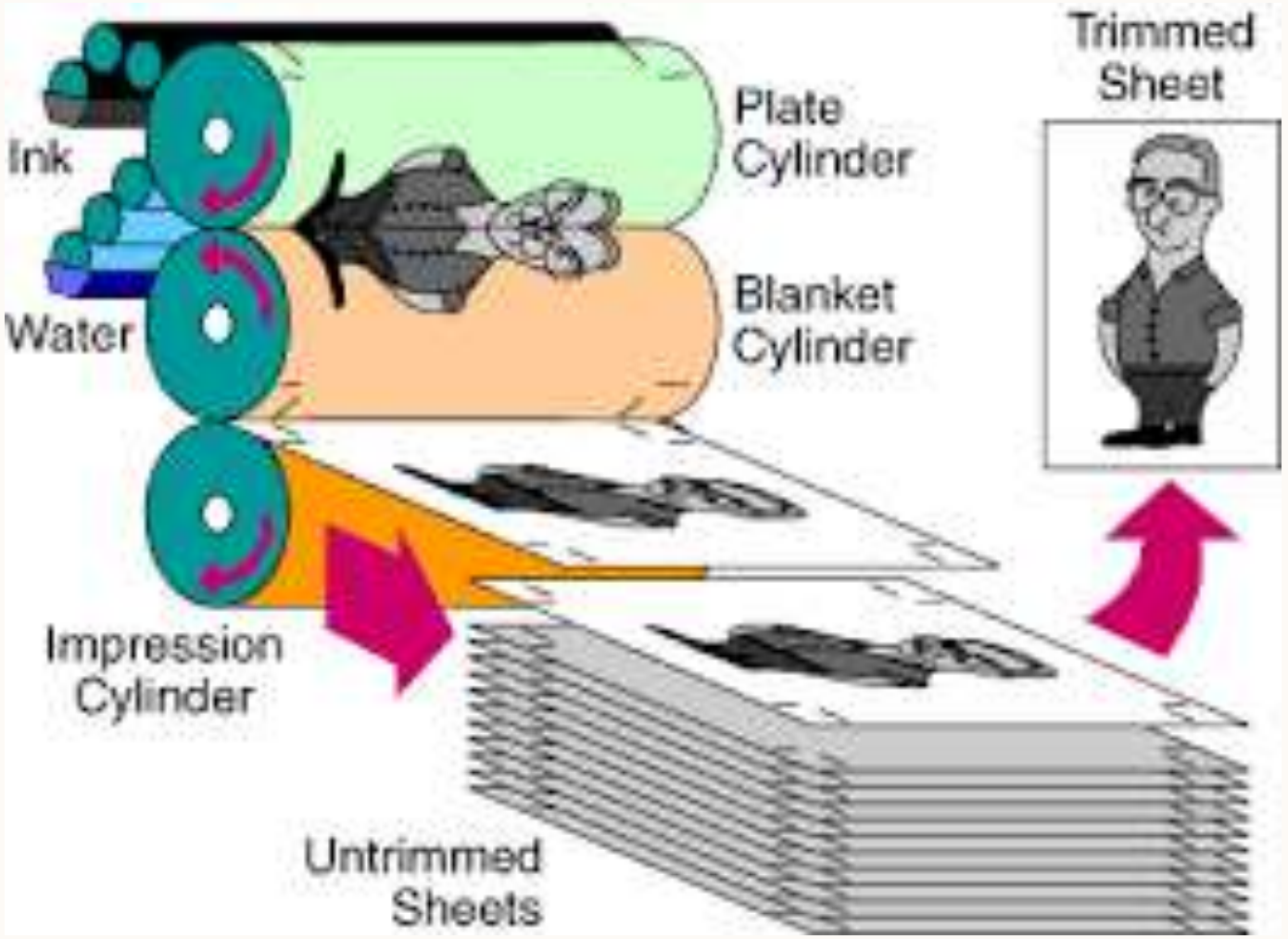
also called offset lithography, is a method of mass-production printing in which the images on metal plates are transferred (offset) to rubber blankets or rollers and then to the print media. This indirect method, based on the principle that water and oil repel each other



The print media, usually paper, does not come into direct contact with the metal plates. This prolongs the life of the plates. In addition, the flexible rubber conforms readily to the print media surface, allowing the process to be used effectively on rough-surfaced media such as canvas, cloth or wood.



- The main advantage of offset printing is its high and consistent image quality.
- The process can be used for small, medium or high-volume jobs.




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- There are two types of offset printing machines in common use for publication today: sheet-fed offset printing and Web offset printing.



SHEET-FED OFFSET PRINTING

- In sheet-fed offset printing, individual pages of paper are fed into the machine. The pages can be pre-cut to the final publication size or trimmed after printing.



- Unlike Web-fed, sheet-fed offset lithography produces smaller quantities as sheets of paper are fed individually to the machine. However, this method is still considered a quick printing option compared to other alternatives.
- Sheet-fed printing can produce up to 12,000 sheets to 24,000 per hour, but with modernization, this type of printing has significantly improved in speed, allowing it to perform similar functions as the web-fed press.

WEB OFFSET PRINTING

- In web offset printing, larger, higher-speed machines are used. These are fed with large rolls of paper and the individual pages are separated and trimmed afterwards. Sheet-fed offset printing is popular for small and medium-sized fixed jobs such as limited-edition books. Web offset printing is more cost-effective for high-volume publications whose content changes often, such as metropolitan newspapers.

- Web-fed offset printing presses can produce over 3,000 papers per minute, w translating to about 80,000 identical sheets per hour, which makes it an ideal option for organizations that need to have large quantities at a fast pace to meet tight deadlines.
- Web offset lithography is typical in the printing of newspapers and boxes. In this process, rolls of paper are fed continuously through the machine press and can produce over 5,000 printed materials. In addition, the paper is cut to create identical sheets during this process.

How It Works

Image Plate:

- A metal plate is etched with the desired image or design.

Ink and Water:

- The plate is first dampened with water, which adheres to the hydrophilic (water-attracting) areas. Then, oil-based ink is applied, adhering only to the hydrophobic (water-repelling) areas that contain the image.

Blanket Transfer:

- The inked image is transferred from the plate to a rubber blanket cylinder.

Printing:

- The rubber blanket cylinder then "offsets" the image onto the final printing surface, such as paper or cardboard, which is held by an impression cylinder.

THE ADVANTAGES AND DISADVANTAGES OF MODERN OFFSET PRINTING

ADVANTAGES:

- High Quality: This process produces high-quality images and text with excellent color accuracy and sharpness, making it an ideal choice for packaging projects that require a professional and eye-catching appearance.
- Versatility: It can be used on various surfaces, including paper, cardboard, and plastic, making it a versatile printing technique for different types of packaging.
- Cost-Effective: Litho-printing is cost-effective for large-scale production runs and has lower setup costs than other printing methods.
- Consistency: Offset printing produces consistent results across large production runs, ensuring that the final product meets the desired quality standards.

DISADVANTAGES:

- **Time-Consuming:** Printing can be time-consuming, requiring several steps, including preparing printing plates.
- **Limited Customization:** While offset printing is versatile, there may be better options for highly customized packaging designs. It may require additional setup time and costs to create custom printing plates.
- **Not Ideal for Small Runs:** Litho-printing is not cost-effective for small production runs, as the setup costs can be high compared to other printing methods.
- **Environmental Concerns:** Using chemicals in the printing process can have ecological implications. It may require proper disposal and handling to minimize their impact.


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QUESTIONS?